

The Two-Angle Admissibility Theorem

Dyadic Radix and Real-Cyclotomic Cosines Force Exactly Two Digit-Extraction Angles

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NEXUS Framework — BBP / Kernel-Frame line

Abstract

A harmonic digit-extraction frame of order n is the pole structure of the kernel $1 - t^n$, whose zero set is the n -th roots of unity. Positional digit extraction requires an integration endpoint u that is simultaneously a mark coordinate of the frame ($u = \cos \vartheta$ for a frame angle ϑ) and a half-integer power of an integer radix ($u^n = 1/b$, $b \in \mathbb{Z}$). This paper proves that these two conditions, taken together, admit **exactly two angles: $\theta = \pi/3$ and $\theta = \pi/4$** . The proof is an intersection statement between a real-cyclotomic field and the rational powers of 2. The key object is $x = 2 \cos \vartheta = \zeta_n + \zeta_n^{-1}$, an algebraic integer; the radix condition forces $x^n = 2^{n-k}$, hence x is a dyadic rational power of 2; and the abelian nature of the maximal real cyclotomic field forces that power to have denominator at most 2. The two special right-triangle cosines survive. The minimal frames realizing the two angles are $(b, n) = (64, 6)$ and $(16, 8)$; every admissible frame lies on one of two towers rooted at these minima. We record what the theorem does *not* establish — the formal necessity of the common readout residue $[4, -2, -1, -1]$ — as the isolated open question the result now exposes.

1. Setup and the Two Endpoint Conditions

Definition 1 (Frame). The frame of order n is the set of directions $F_n = \{e^{2\pi i j/n} : 0 \leq j < n\}$, the zero set of $1 - t^n$. A *mark coordinate* is a real part $\cos(2\pi j/n)$ of a mark.

Definition 2 (Shell constant). For frame order n , radix b , and endpoint u with $u^n = 1/b$, the shell constant of address r ($r = 1, \dots, n$) is

$$C_r = \sum_{k \geq 0} 1 / (b^k (n k + r)) = u^{-r} \int_0^u t^{r-1} / (1 - t^n) dt.$$

The second equality (proved in the companion Kernel-Frame paper) is the bridge we use: any constant relation $\sum_r a_r C_r = o$ is equivalent to a numerator polynomial $N(t) = \sum_r a_r u^{-r} t^{r-1}$ with $\int_0^u N(t)/(1-t^n) dt = o$.

The two admissibility conditions. Positional digit extraction (base- b read-out) requires the endpoint to be, simultaneously:

(A) a mark coordinate: $u = \cos \vartheta$ for a frame angle $\theta = 2\pi j/n$;

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(R) a dyadic-radix power: $u^n = 1/b$ with π -producing integer $b \geq 2$.

Condition (A) is frame geometry; condition (R) is radix arithmetic. The theorem below is the intersection of the two. We take $u > 0$ (a nondegenerate positive endpoint) throughout, and treat the full-turn case $u = 1$ separately as degenerate.

2. The Radix Is Forced Dyadic

Lemma 1 (Dyadic radix). Conditions (A) and (R) together imply $b = 2^k$ for some integer $0 \leq k \leq n$.

Proof. Set $x = 2 \cos \theta$. Since $\theta = 2\pi j/n$ is a rational multiple of π , $x = \zeta_n + \zeta_n^{-1}$ where $\zeta_n = e^{2\pi i/n}$. A sum of a root of unity and its inverse is an *algebraic integer* (it is a root of a monic integer polynomial — the minimal polynomial of $2\cos(2\pi j/n)$ over $\mathbb{Q}$ is monic with integer coefficients). It lies in the maximal real cyclotomic field $\mathbb{Q}(\zeta_n)^+ = \mathbb{Q}(\zeta_n + \zeta_n^{-1})$.

Caution (why x , not u). The algebraic-integer property belongs to $x = 2 \cos \theta$, **not** to $u = \cos \theta$. For the hexagonal seed $u = 1/2$ is a rational non-integer; $x = 1$ is the algebraic integer. Stating the step on u would be false. This is the single correction that makes the spine rigorous.

Now combine with (R). Since $u = x/2$,

$$x^n = (2u)^n = 2^n \cdot u^n = 2^n / b.$$

The left side x^n is an algebraic integer (a power of one). The right side $2^n/b$ is rational. A rational algebraic integer is a rational integer, so

$$2^n / b \in \mathbb{Z} \implies b \mid 2^n \implies b = 2^k, 0 \leq k \leq n. \blacksquare$$

3. The Endpoint Is a Dyadic Power of Bounded Degree

Lemma 2 (Bounded degree). Under (A) and (R), $x = 2 \cos \theta$ equals a rational power $2^{a/d}$ with $\gcd(a, d) = 1$ and $d \leq 2$.

Proof. By Lemma 1, $x^n = 2^{n-k}$, so $x = 2^{(n-k)/n}$. Write the exponent in lowest terms, $(n-k)/n = a/d$ with $\gcd(a, d) = 1$, so $x = 2^{a/d}$. For $a > 0$, $\mathbb{Q}(2^{a/d}) = \mathbb{Q}(2^{1/d})$, and $2^{1/d}$ has degree exactly d over $\mathbb{Q}$: for $d \geq 2$ the polynomial $X^d - 2$ is irreducible by Eisenstein at the prime 2.

The cyclotomic side constrains this degree. By Lemma 1's proof, $x \in \mathbb{Q}(\zeta_n)^+$, and $\mathbb{Q}(\zeta_n)/\mathbb{Q}$ is **abelian** (Galois with group $(\mathbb{Z}/n\mathbb{Z})^\times$), so its subfield $\mathbb{Q}(\zeta_n)^+$ is abelian over $\mathbb{Q}$ as well. Every subfield of an abelian extension is itself Galois over $\mathbb{Q}$ (the corresponding subgroup is normal because every subgroup of an abelian group is normal). In particular $\mathbb{Q}(x) = \mathbb{Q}(2^{1/d})$ must be **Galois over $\mathbb{Q}$** .

But $\mathbb{Q}(2^{1/d})$ is *not* normal for $d \geq 3$: the conjugates $2^{1/d}\zeta_d^j$ for $j \geq 1$ are non-real, so the real field $\mathbb{Q}(2^{1/d})$ cannot contain them and is not closed under the Galois action. A Galois field that is not normal is a contradiction. Hence $d \leq 2$. $\blacksquare$

4. Enumeration: Exactly Two Angles

Theorem (Two-Angle Admissibility). Let F_n be a frame with a positive nondegenerate endpoint $u = \cos \vartheta$ satisfying the dyadic-radix condition $u^n = 1/b$, $b \in \mathbb{Z}_+$. Then

$$\theta = \pi/3 \quad \text{or} \quad \theta = \pi/4,$$

equivalently $\cos \vartheta = 1/2$ or $\cos \vartheta = 1/\sqrt{2}$, i.e. $x = 2 \cos \vartheta = 1$ or $\sqrt{2}$.

Proof. By Lemma 2, $x \in \{2^{a/d} : d \leq 2\}$, and the endpoint constraint gives $0 < x \leq 2$ (since $|\cos \vartheta| \leq 1$). Split by d :

Case $d = 1$ (rational x). Then $x = 2^a$ is an integer power of 2 in the interval $(0, 2]$, so $x \in \{1, 2\}$. By Niven's theorem the only rational values of $\cos$ of a rational-angle are 0, $\pm 1/2$, ± 1 ; the positive ones giving these x are $\cos \theta = 1$ ($x = 2$, the degenerate full turn $\theta = 0$, no frame) and $\cos \theta = 1/2$ ($x = 1$, $\theta = \pi/3$). The value $x = 2^{-1}$ is excluded: it needs $\cos \theta = 1/4$, not a Niven value.

Case $d = 2$ (quadratic x). Then $x = 2^{a/2}$ with a odd (else the power reduces to $d = 1$). From $0 < x \leq 2$ we get $x^2 = 2^a \leq 4$, so $a \leq 2$; with a odd this forces $a = 1$ and $x = \sqrt{2}$. Then $\cos \theta = 1/\sqrt{2}$, i.e. $\theta = \pi/4$. Consistency check: $\mathbb{Q}(\sqrt{2})$ is exactly the real quadratic subfield $\mathbb{Q}(\zeta_8)^+$ of the 8th cyclotomic field — the minimal frame carrying $\pi/4$. ■

Computational confirmation. An independent numeric scan over all frames and mark coordinates $2 \cos(2\pi j/n)$ for $n \leq 200$ returns exactly the dyadic-power values $\{1, \sqrt{2}\}$ (plus the degenerate 2 at $j = 0$), with **no degree- ≥ 3 dyadic power ever occurring** — matching the abelian-subfield prediction of Lemma 2.

5. Frame Minima and the Two Towers

The two admissible angles have minimal frames determined by $b = 2^k$ with k chosen so the endpoint mark is the radix power:

Angle θ	$\cos \theta$	$x = 2 \cos \theta$	Order n	Radix b
$\pi/3$	$1/2$	1	6	$2^6 = 64$
$\pi/4$	$1/\sqrt{2}$	$\sqrt{2}$	8	$2^4 = 16$

Not two pairs — two towers. Each angle generates a tower of admissible frames, one per multiple of the base order, with the endpoint held fixed at its mark value and the radix rising accordingly:

hexagonal ($\theta = \pi/3$): $(64, 6), (2^{12}, 12), (2^{18}, 18), \dots$ $b = 2^n$, endpoint $1/2$

octal ($\theta = \pi/4$): $(16, 8), (2^8, 16), (2^{12}, 24), \dots$ $b = 2^n$, endpoint $\cos(\pi/4)$

The pullback map $b - t^n \mapsto b - t^{kn}$ transports the kernel (verified: rank $K = 1$ preserved, null vector transported by zero-interleaving under $t \mapsto t^k$), but does *not* preserve the endpoint as a frame mark — the pullback member $(64, 12)$ has endpoint $2^{-1/2}$, which is off the F_{12} grid. The admissible rung is $(2^{12}, 12)$, not $(64, 12)$. This is why **admissible angle does not imply a read-out defect at every (b, n) pair** — only along the correctly-radixed tower.

6. Status: What This Establishes, and the Isolated Open Question

The theorem replaces a two-point empirical coincidence with a structural intersection. Several earlier candidate mechanisms are now understood as read-side shadows of the dyadic-radix condition, not causes:

Claim	Status	Basis
$b \mid 2^n$, hence $b = 2^k$	THEOREM	Lemma 1 ($x = 2\cos\theta$ algebraic integer + rational)
Endpoint is a dyadic power $2^{a/d}$, $d \leq 2$	THEOREM	Lemma 2 (abelian subfield of $\mathbb{Q}(\zeta_n)^+$)
Exactly two angles $\pi/3$, $\pi/4$	THEOREM	Section 4 enumeration + Niven
Frame minima $(64,6)$, $(16,8)$	THEOREM	Section 5
Tower kernel transport, $K = 1$ at each rung	VERIFIED	PSLQ, norms 933 & 217 preserved
$[4, -2, -1, -1]$ survives up each tower	VERIFIED	residue = winding-sum of stencil, scaled by b-powers
Mersenne / $k \mid n$ signature	⊥ SHADOW	$b = 2^k$ derived; $k \mid n$ a partial correlate
Cyclotomic-field / degree-alone criteria	⊥ SHADOW	same field, same n occur without defect
Formal necessity of residue $[4, -2, -1, -1]$	Ω OPEN	the next load-bearing question

The open question, isolated. The classification of admissible *frames* is now closed to two towers. What remains is a *readout* question of a different type: the common residue $[4, -2, -1, -1]$ is **verified** to be the primitive readout stencil of both towers — identical across the hexagonal and octal families, invariant under pullback up each tower — but it is not yet **derived** as a formal consequence of the two admissible angles. The precise open statement is:

Ω: Does the common residue $[4, -2, -1, -1]$ follow formally from $\theta \in \{\pi/3, \pi/4\}$,

or is it an independent readout fact the surviving frames merely share?

This is a sharper and better-posed problem than the empirical cross-angle comparison of the two towers' π -observables ($4\pi/3$ and $4\pi/\sqrt{3}$ hexagonal versus π octal), which is downstream: once the residue's status is settled, the cross-angle relation becomes a consequence question rather than another search.

Layer discipline maintained throughout: radix → endpoint → admissible angle → frame → residue is one channel; frame → kerΦ is a separate channel. Pullback acts on the second without guaranteeing the first. The residue is the invariant readout; the null vector is frame-specific and changes up the tower.

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